The Role of Weather in French Day-Ahead Electricity Price Forecasting: A Random Forest Approach

Master's Thesis

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Abstract

This thesis investigates whether weather variables significantly improve day-ahead electricity price forecasting for the French market (EPEX SPOT France) using machine learning models. We build a comprehensive dataset spanning January 2018 to April 2025, combining hourly price and generation data from the ENTSO-E Transparency Platform with ERA5 reanalysis weather data, cross-border flow data, and fuel price series (TTF natural gas, ARA coal, EU ETS carbon allowances).

Four models are compared: a naïve lag-168h benchmark (Model A), a Random Forest without weather features (Model B), a Random Forest with full weather features (Model C), and an XGBoost model with weather features (Model D). Models are evaluated on a strict out-of-sample test set covering May 2024 to April 2025 (8,640 hours).

Our central finding is that weather features do not significantly improve forecast accuracy over the enriched fundamental model, as confirmed by a Diebold-Mariano test ($p = 0.572$). We attribute this to the fact that fuel prices (TTF gas) and nuclear availability already implicitly encode meteorological information in the French market context. Both Random Forest models achieve a Mean Absolute Error of approximately 16.9 EUR/MWh and $R^2 \approx 0.776$, representing a 49% reduction in error over the naïve benchmark. A day-ahead directional trading backtest confirms the economic value of these forecasts, with RF models generating net profits of approximately 136,700 EUR/MW over the test period with a maximum drawdown ten times smaller than the naïve strategy.

Keywords: electricity price forecasting, day-ahead market, EPEX SPOT France, Random Forest, XGBoost, weather features, Diebold-Mariano test, feature engineering, trading backtest.

Acknowledgements

We would like to thank the EDHEC MSc Data Analysis & AI faculty for their guidance throughout this project. We are grateful to RTE (Réseau de Transport d'Électricité) and ENTSO-E for making electricity market data publicly available through the Transparency Platform, and to the Copernicus Climate Change Service for providing open access to ERA5 reanalysis data.

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List of Acronyms

DA Day-Ahead

DM Diebold-Mariano

ENTSO-E European Network of Transmission System Operators for Electricity

ETS Emissions Trading System

EUA European Union Allowance

HDD Heating Degree Days

CDD Cooling Degree Days

CSS Clean Spark Spread

CDS Clean Dark Spread

MAE Mean Absolute Error

RMSE Root Mean Square Error

sMAPE Symmetric Mean Absolute Percentage Error

MDD Maximum Drawdown

MDI Mean Decrease in Impurity

RF Random Forest

RTE Réseau de Transport d'Électricité

TSO Transmission System Operator

WSI Weather Stress Index

Chapter 1

Introduction

1.1 Context and Motivation

Electricity markets have undergone a fundamental transformation over the past three decades. The progressive liberalisation of European energy markets, accelerated by successive EU directives from 1996 onwards, replaced vertically integrated state monopolies with competitive wholesale markets in which prices are determined by the interaction of supply and demand. In France, the day-ahead electricity market operates through EPEX SPOT, where participants submit bids and offers for each hour of the following day, with a single clearing price determined by a batch auction that closes at 12:00 CET.

The resulting price series is among the most complex in quantitative finance. Unlike commodity prices, electricity cannot be economically stored at scale, which forces supply and demand into instantaneous equilibrium at every moment. The consequence is a price process characterised by extreme volatility, sharp intraday seasonality, strong weekly periodicity, and irregular price spikes that can reach several thousand euros per megawatt-hour during scarcity episodes. The French market adds a layer of specificity: with approximately 63 GW of installed nuclear capacity representing roughly 70% of annual electricity production (RTE, 2023), the availability of the nuclear fleet is a primary determinant of price levels. In parallel, France's residential heating stock is predominantly electric, creating a thermosensitivity of approximately 2.4 GW per degree Celsius in winter (RTE, 2024) — the highest in Europe.

Accurate forecasting of day-ahead prices is therefore not merely an academic exercise. It is a critical input for energy suppliers setting customer tariffs, for industrial consumers optimising consumption schedules, for renewables producers valuing flexibility options, and for traders positioning in forward markets. A one euro per megawatt-hour improvement in forecast accuracy over a 100 MW portfolio translates to approximately 876,000 EUR in annual value capture — motivating sustained interest in more accurate models.

1.2 Research Question

This thesis addresses the following central question:

Do weather variables significantly improve the accuracy of day-ahead elec-

tricity price forecasts for the French market, and if so, what is the economic value of that improvement?

This question is decomposed into three sub-questions:

1. Can machine learning models (Random Forest, XGBoost) substantially outperform a naïve statistical benchmark on French day-ahead prices?
2. Does the addition of meteorological features (temperature, wind speed, solar radiation, Heating Degree Days, Weather Stress Index) produce a statistically significant improvement over a model built on market fundamentals alone?
3. Can the predictive improvement of these models be translated into economically meaningful returns through a day-ahead directional trading strategy?

1.3 Contribution

This thesis makes three contributions to the electricity price forecasting literature:

- **Comprehensive French dataset:** We assemble and publish a reproducible dataset spanning 2018–2025 that combines ENTSO-E generation and load data, ERA5 reanalysis weather data, cross-border flow data, and fuel price series. To our knowledge, this is one of the most complete publicly-sourced datasets for the French day-ahead market.
- **Rigorous ablation study:** We design a four-model experiment (A/B/C/D) that isolates the marginal contribution of weather features using a Diebold-Mariano test with Harvey-Leybourne-Newbold correction. The result — that weather does not significantly improve forecast accuracy in the enriched fundamental model — is a non-trivial finding that contributes to the understanding of information redundancy in electricity price drivers.
- **Day-ahead executable trading backtest:** We implement a directional trading strategy using an economically coherent signal design grounded in the actual mechanics of the EPEX SPOT day-ahead auction, with realistic transaction cost scenarios drawn from the 2024 EPEX fee schedule.

1.4 Structure of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 reviews the literature on electricity price forecasting, covering both statistical and machine learning approaches. Chapter 3 describes the data sources, feature engineering process, and exploratory

analysis. Chapter 4 presents the modelling framework, results, and statistical tests. Chapter 5 introduces the trading backtest and its results. Chapter 6 discusses the findings, limitations, and extensions. Chapter 7 concludes.

All code, data pipelines, trained models, and figures are available in the project repository at https://github.com/leogthub/fr-power-thesis.

Chapter 2

Literature Review

2.1 Overview of Electricity Price Forecasting

The electricity price forecasting literature has expanded rapidly since market liberalisation in the 1990s. Weron (2014) provides the definitive survey, classifying approaches into five families: (1) multi-agent models, (2) fundamental models, (3) reduced-form stochastic models, (4) statistical models, and (5) computational intelligence models. More recently, Nowotarski and Weron (2018) document the shift towards probabilistic forecasting and machine learning methods. For the purposes of this thesis, we focus on short-term point forecasting at the day-ahead horizon, which remains the most practically relevant setting for market participants.

2.2 Statistical and Econometric Approaches

Early work on day-ahead price forecasting relied on time-series models adapted from financial econometrics. Autoregressive models — in particular ARIMA and its seasonal extensions — capture the strong autocorrelation structure of electricity prices but struggle to model the non-linear interactions between price and its fundamental drivers. GARCH-family models address volatility clustering but are ill-suited to the extreme price spikes that characterise electricity markets, where the distribution of returns has much heavier tails than financial asset returns.

The mean-reversion and jump-diffusion framework, introduced in the commodity pricing literature, is better able to capture the spike-and-reversion behaviour of electricity prices (Fanone et al., 2013). However, these models typically require numerical methods for calibration and are difficult to extend to include exogenous weather or fundamental variables.

Factor models represent a middle ground. Maciejowska et al. (2016) apply Factor Quantile Regression Averaging to day-ahead prices across European markets, combining multiple linear specifications through optimal weights. Uniejewski et al. (2019) propose automated variable selection using LASSO regularisation applied to a rich set of price lags and calendar variables, achieving competitive performance relative to more complex non-linear models. This result — that well-specified linear models with price lags can be surprisingly competitive — is relevant to our own findings.

2.3 Machine Learning Approaches

The application of machine learning to electricity price forecasting has accelerated since the mid-2010s, driven by the increasing availability of high-resolution data and cheap computational resources. Aggarwal et al. (2009) provide an early review, while Lago et al. (2018) offer a more recent empirical comparison of deep learning architectures (LSTM, CNN) against traditional benchmarks on European day-ahead markets.

Tree-based ensemble methods — Random Forests (Breiman, 2001) and Gradient Boosted Trees (including XGBoost, Chen and Guestrin 2016) — have proven particularly well-suited to the electricity price forecasting problem for several reasons. First, they handle non-linear interactions between weather, calendar, and fundamental variables without requiring manual feature transformations. Second, they are robust to outliers and near-zero prices, which are frequent in liberalised markets and create difficulties for methods that rely on symmetric loss functions. Third, tree ensembles provide built-in feature importance measures (Mean Decrease in Impurity for Random Forests) that offer interpretable insights into price drivers, a property highly valued by market participants who must justify their models to risk committees.

Ziel and Weron (2018) compare univariate and multivariate forecasting frameworks on European day-ahead markets and find that multivariate models incorporating load, generation, and cross-border flows outperform univariate price-only models. This supports the use of fundamental variables in our feature matrix. However, none of the reviewed studies specifically test the marginal contribution of weather variables in the French market context using formal statistical tests.

2.4 Weather as a Driver of Electricity Prices

The causal pathway from weather to electricity prices is well-established in the fundamental literature. Temperature affects demand through heating (in winter) and cooling (in summer). Wind speed determines the output of wind generators, which set the marginal cost to zero when in excess, driving prices down. Solar irradiance similarly compresses midday prices during high-penetration periods. Precipitation affects hydro reservoir levels, with droughts leading to higher thermal generation and higher prices.

Staffell and Pfenninger (2016) document the strong correlation between wind speed and wind generation output (approximately cubic relationship), which underlies our *wind power proxy* feature. Lindberg et al. (2019) examine the interaction between wind and hydro in Nordic markets, where weather has a particularly dominant role.

In the French market specifically, the thermosensitivity is unusually high by European standards. RTE (2024) report that each degree Celsius below 17°C increases peak demand by approximately 2.4 GW — a consequence of the high penetration of electric

heating (over 30% of French households). This motivates the use of 17°C as the Heating Degree Day threshold throughout this thesis, consistent with RTE's own operational methodology.

However, the relationship between weather and price is not purely additive. Cold temperatures combined with low wind speed — a combination we call the *Weather Stress Index* — are particularly dangerous for the French system, as they simultaneously increase demand and reduce renewable generation. Such episodes have historically driven the most extreme price spikes. Our WSI feature, defined as HDD divided by wind speed, attempts to capture this interaction.

2.5 Diebold-Mariano Testing in Energy Forecasting

Comparing forecast accuracy across models requires careful statistical treatment. Informal comparisons based on point estimates of MAE or RMSE can lead to misleading conclusions when differences are not statistically significant. The Diebold-Mariano (DM) test (Diebold and Mariano, 1995) provides a formal framework for testing the null hypothesis of equal predictive accuracy between two competing forecasts. The test statistic is based on the difference in loss functions (here, absolute errors) and accounts for autocorrelation in the loss differential series via a Newey-West long-run variance estimator (Newey and West, 1987).

For small samples, Harvey et al. (1997) propose a correction factor that adjusts the DM statistic to better approximate a t-distribution, reducing the risk of over-rejection of the null. We apply this Harvey-Leybourne-Newbold (HLN) correction throughout our analysis, following best practice in the forecasting literature (Weron, 2014).

2.6 Gap in the Literature

Despite the growing body of work on machine learning for electricity price forecasting, several gaps remain relevant to this thesis:

1. **French market specificity:** Most empirical studies focus on the German, Spanish, or Nordic markets. The French market's combination of high nuclear share, high thermosensitivity, and interconnection with multiple neighbours creates a distinct price formation mechanism that deserves dedicated analysis.

2. **Formal ablation of weather features:** While weather variables are routinely included in forecasting models, their marginal contribution is rarely isolated using a statistically formal test. We address this gap with a Diebold-Mariano ablation study.

3. **Economic value evaluation:** Most studies evaluate models on statistical metrics alone. We extend the analysis to an executable day-ahead trading backtest with realistic transaction costs, linking statistical accuracy to economic value.

Chapter 3

Data and Feature Engineering

3.1 Data Sources

Our dataset combines four primary sources, covering the period January 2018 to April 2025 at hourly resolution, yielding approximately 64,000 observations before feature construction.

3.1.1 ENTSO-E Transparency Platform

The European Network of Transmission System Operators for Electricity (ENTSO-E) makes hourly data publicly available through its Transparency Platform (ENTSO-E, 2024). We retrieve the following series for France (bidding zone `10YFR-RTE---C`):

- **Day-ahead prices** (EUR/MWh): the clearing price of the EPEX SPOT France day-ahead auction, which constitutes our forecasting target.
- **Actual total load** and **load forecast** (MW): the TSO's hourly load forecast, published the day before delivery, serves as a key predictor of demand pressure.
- **Actual generation by source** (MW): nuclear, gas, run-of-river hydro, reservoir hydro, solar, and onshore wind.
- **Cross-border physical flows** (MW): net hourly flows between France and its four main interconnected neighbours — Germany, Spain, the United Kingdom, and Belgium. These flows reflect relative price differentials and capacity constraints across the European internal energy market.

Data are retrieved programmatically using the `entsoe-py` Python library with the official ENTSO-E API key. All timestamps are converted to UTC+1 (Europe/Paris) and aligned on a consistent hourly index.

3.1.2 ERA5 Reanalysis Weather Data

Meteorological variables are sourced from the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) through the Copernicus Climate Change Service (Hersbach et al., 2020). ERA5 provides hourly estimates of atmospheric variables on a $0.25° \times 0.25°$ grid, covering the globe from 1940 to present.

We retrieve the following variables for a bounding box covering metropolitan France (41–51°N, -5–10°E) and compute spatial means weighted by grid area:

- 2-metre air temperature (°C)
- 10-metre wind speed (m/s)
- Surface solar irradiance (W/m^2, derived from surface net solar radiation)
- Total precipitation (m/h)

Data are downloaded month-by-month using the `cdsapi` library to avoid request size limitations, then concatenated and temporally aligned with the ENTSO-E series.

3.1.3 Fuel Prices

Fuel prices are retrieved from public market data sources:

- **TTF natural gas** (EUR/MWh): the Title Transfer Facility day-ahead price, the European gas benchmark. TTF prices are the primary determinant of the marginal cost of gas-fired generation, which is frequently the price-setting technology in France outside nuclear baseload hours.
- **ARA coal** (EUR/tonne): the price of thermal coal delivered to Amsterdam-Rotterdam-Antwerp. Together with EU ETS carbon allowance prices, coal prices determine the Clean Dark Spread and the competitiveness of coal against gas.

3.1.4 Price Statistics

Figure 3.1 shows the full price series from 2018 to 2025. The period exhibits three distinct regimes: a pre-crisis period (2018–2021) with prices typically between 20–80 EUR/MWh; the energy crisis of 2021–2023, during which French prices reached historical extremes above 1,000 EUR/MWh driven by the simultaneous Russian gas supply disruption and the historically low nuclear availability due to corrosion-related maintenance shutdowns; and a normalisation phase from 2023 onwards as gas prices declined and nuclear availability recovered toward 75–80%.

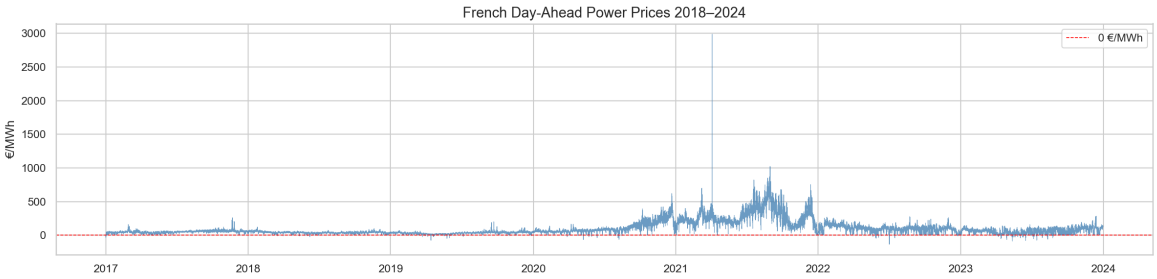

Figure 3.1: French day-ahead electricity prices (EUR/MWh), January 2018 – April 2025. The 2022 energy crisis produced historically extreme prices, driven by the simultaneous gas supply shock and low nuclear availability.

The distribution of hourly prices is heavily right-skewed with a non-trivial mass of negative observations (Figure 3.2). Negative prices occur when inflexible generators (nuclear, wind) are constrained to run and cannot be economically curtailed, resulting in producers paying consumers to absorb excess generation.

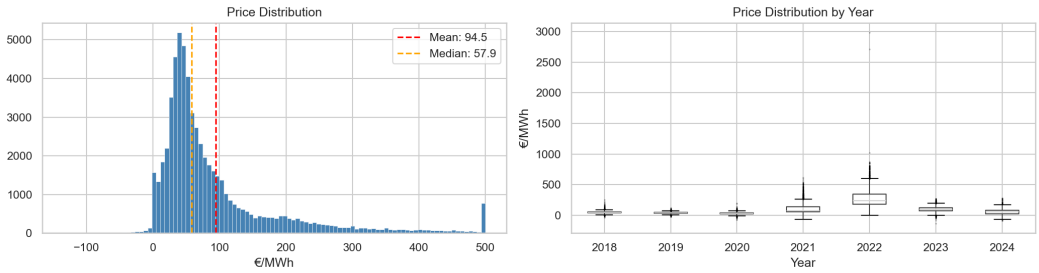

Figure 3.2: Distribution of hourly day-ahead prices (left) and proportion of negative-price hours by year (right). Negative prices account for approximately 3–8% of hours in recent years.

The strong seasonality at multiple frequencies is apparent in Figure 3.3, which shows the average price by hour of day and month of year. Winter morning and evening peaks, summer midday price compressions from solar generation, and the consistent week/weekend differential are all visible.

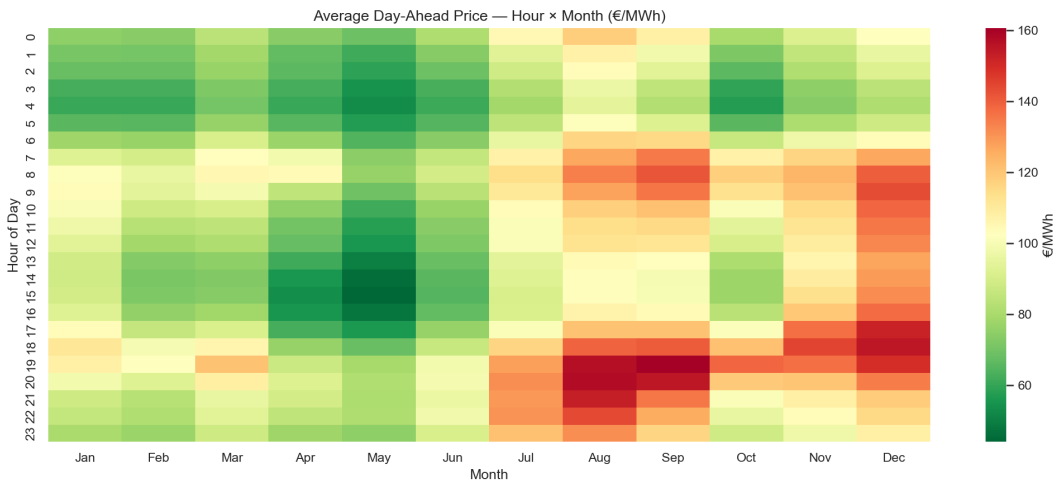

Figure 3.3: Average hourly price by hour of day and month of year (2018–2025). Dark shading indicates high prices; winter peak hours (07:00–09:00, 18:00–20:00) are systematically elevated.

3.2 Feature Engineering

From the raw data sources described above, we construct 35 features organised into five categories. The full list is provided in Appendix B; we describe the most methodologically significant features here.

3.2.1 Calendar Features

Calendar variables capture the deterministic seasonality of electricity demand. We encode the hour of day and month of year using sine/cosine transformations to preserve their cyclic nature:

$$\text{hour_sin} = \sin\left(\frac{2\pi \cdot h}{24}\right), \quad \text{hour_cos} = \cos\left(\frac{2\pi \cdot h}{24}\right) \tag{3.1}$$

This encoding ensures that the model correctly identifies hour 23 and hour 0 as adjacent, which a linear encoding would fail to capture. An `is_weekend` binary indicator captures the systematic weekday/weekend price differential visible in Figure 3.3.

3.2.2 Price Lags

Price lags are the most informative predictors in our feature matrix, reflecting the strong autocorrelation structure of electricity prices. We include lags at 24h, 48h, and 168h (one week), as well as 24-hour and 168-hour rolling means computed on the lagged series to prevent data leakage:

$$\bar{p}_{t,24} = \frac{1}{24} \sum_{k=25}^{48} p_{t-k} \tag{3.2}$$

The 168-hour lag (same hour, previous week) is particularly important: weekly seasonality in electricity prices is strong, and the previous week's same-hour price is a better predictor than the previous day's price in many market conditions.

3.2.3 Weather-Derived Features

Beyond the raw ERA5 variables, we construct three derived features specifically motivated by the French market context:

Heating Degree Days (HDD):

$$\text{HDD}_t = \max(0,\ 17 - T_t) \tag{3.3}$$

where T_t is the spatially-averaged 2m temperature at hour t. The threshold of 17°C is the standard used by RTE for French demand modelling (RTE, 2024), reflecting the predominantly electric heating stock. This is higher than the 15°C threshold commonly used in Germany, consistent with France's stronger thermosensitivity.

Wind Power Proxy:

$$\text{wind_proxy}_t = v_t^3 \tag{3.4}$$

where v_t is the 10m wind speed. Wind turbine power output is proportional to the cube of wind speed in the operational range, making this a physically motivated transformation (Staffell and Pfenninger, 2016).

Weather Stress Index (WSI):

$$\text{WSI}_t = \frac{\text{HDD}_t}{v_t + \varepsilon} \tag{3.5}$$

where $\varepsilon = 0.1$ prevents division by zero. The WSI is high when temperatures are cold AND wind is weak — the combination that maximises electricity demand while minimising wind generation, historically associated with the most extreme price spikes in France.

3.2.4 Nuclear Availability Ratio

$$\text{nuclear_avail}_t = \frac{\text{gen_nuclear}_t}{63{,}000\ \text{MW}} \tag{3.6}$$

This ratio captures the fraction of France's installed nuclear fleet currently online (RTE, 2023). A ratio below 0.7 signals significant outage periods and is historically associated with higher prices, as illustrated in Figure 3.4.

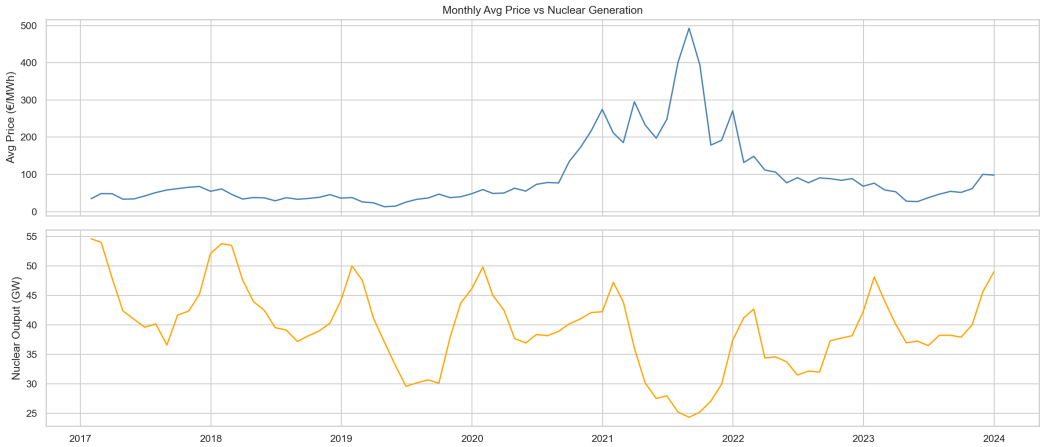

Figure 3.4: Monthly average day-ahead price vs. nuclear availability ratio (2018–2025). The negative correlation is particularly strong during the 2022 outage crisis when availability fell below 0.5.

3.2.5 Weather and Price Correlation

Figure 3.5 shows the relationship between temperature and day-ahead price. The expected U-shaped pattern is visible: prices rise at very low temperatures (heating demand) and slightly at very high temperatures (cooling demand), with a trough around 15–20°C. Figure 3.6 illustrates the Weather Stress Index seasonality, with peaks in January–February coinciding with maximum heating demand and minimum wind generation probability.

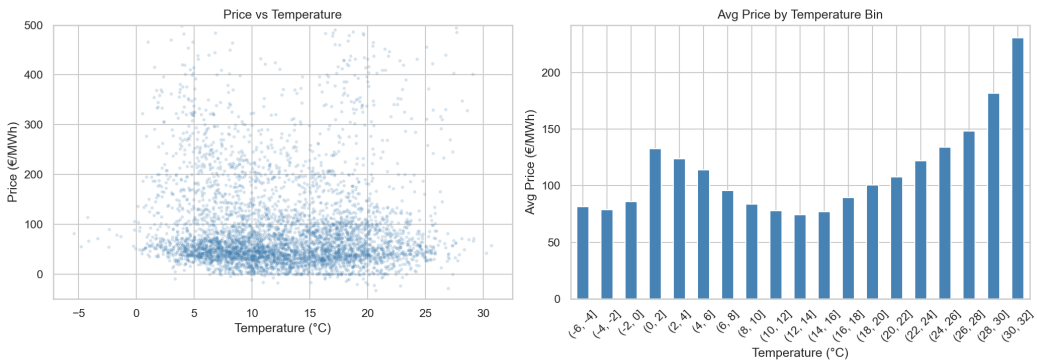

Figure 3.5: Day-ahead price vs. 2m temperature (hexbin density plot). The non-linear U-shaped relationship motivates the use of HDD/CDD rather than raw temperature.

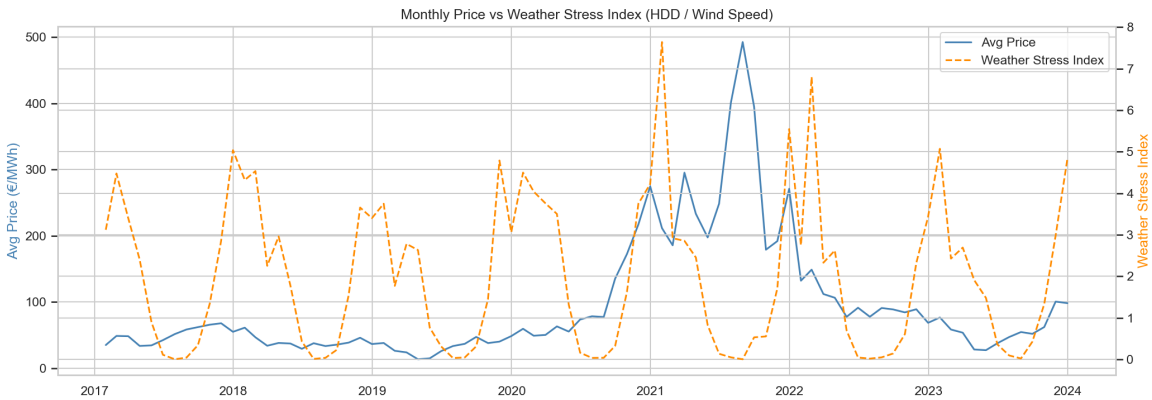

Figure 3.6: Weather Stress Index (WSI) seasonal profile and correlation with day-ahead prices. High WSI values are concentrated in December–February and are associated with elevated price volatility.

3.3 Train / Test Split

A strict temporal split is applied to prevent data leakage. The model is trained on all hours from January 2018 to April 2024, and evaluated on the out-of-sample test set from May 2024 to April 2025 (12 months, 8,640 hours). No information from the test period is used in feature construction, model selection, or hyperparameter tuning.

Table 3.1: Train and test set summary

Set	Period	Hours	Share
Training	Jan 2018 – Apr 2024	55,248	86.4%
Test (out-of-sample)	May 2024 – Apr 2025	8,640	13.6%
Total	Jan 2018 – Apr 2025	63,888	100%

The test period (May 2024 – April 2025) is a "normal" market regime characterised by moderate price levels (30–80 EUR/MWh), recovered nuclear availability (75–80%), and declining gas prices. This contrasts with the training set, which includes the extreme 2022 crisis. The model must therefore generalise from a period of extreme stress to a period of normalisation — a realistic and demanding out-of-sample challenge.

Data cleaning: Columns with more than 50% missing values are dropped. Remaining gaps (typically up to 3 consecutive hours) are filled by forward-fill (`ffill`). Rows where the target variable or core lag features are missing are dropped. This procedure results in 63,888 clean observations.

Chapter 4

Modelling

4.1 Model Architecture

We estimate four models that form a nested hierarchy, allowing us to decompose the sources of forecast improvement:

Model A — Naïve benchmark The naïve forecast assigns to each hour of the test set the realised price from the same hour of the previous week:

$$\hat{p}_t^A = p_{t-168} \tag{4.1}$$

This lag-168h benchmark captures the strong weekly seasonality of electricity prices and is more demanding than the commonly used lag-24h naive, as weekly patterns are more regular than daily ones. It provides the minimum performance bar that any predictive model must exceed.

Model B — Random Forest without weather A Random Forest regressor trained on 27 features: calendar variables, price lags, generation fundamentals, cross-border flows, and fuel prices. All ERA5 meteorological variables and derived weather features (HDD, CDD, wind power proxy, WSI) are excluded. Model B serves as the ablation baseline: its comparison to Model C isolates the marginal contribution of weather.

Model C — Random Forest with weather (main model) An identical Random Forest trained on the full 35-feature matrix, including all meteorological variables. This is our primary model.

Model D — XGBoost with weather An XGBoost gradient-boosted tree model trained on the same 35-feature matrix as Model C. Its comparison to Model C assesses whether a more expressive boosted model can improve over the Random Forest.

4.2 Random Forest

A Random Forest (Breiman, 2001) is an ensemble of B decision trees, each trained on a bootstrap sample of the training data with a random subset of m features considered

at each split. The ensemble prediction is the average across all trees:

$$\hat{p}_t^{\mathrm{RF}} = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}_t) \tag{4.2}$$

The randomisation at both the observation and feature level reduces variance relative to a single deep tree without introducing the high bias of shallow trees. For our application, the following hyperparameters are used (see Appendix A): $B = 500$ trees, maximum depth of 10, minimum of 5 observations per leaf. The `sqrt` feature selection rule is applied at each split.

Random Forests are well-suited to electricity price forecasting for three reasons. First, the tree structure naturally captures threshold non-linearities (e.g., the effect of temperature below the 17°C heating threshold). Second, the ensemble is robust to the outlier prices that are common in electricity markets. Third, the Mean Decrease in Impurity (MDI) importance measure provides model-level interpretability that linear models cannot offer for non-linear interactions.

4.3 XGBoost

XGBoost (Chen and Guestrin, 2016) builds an ensemble of trees sequentially, where each tree corrects the residuals of the previous ensemble:

$$\hat{p}_t^{(k)} = \hat{p}_t^{(k-1)} + \eta \cdot T_k(\mathbf{x}_t) \tag{4.3}$$

where η is the learning rate and T_k is fitted to the negative gradient of the loss function. L1 and L2 regularisation terms on the tree leaf weights prevent overfitting. Our hyperparameters: 500 estimators, learning rate $\eta = 0.05$, maximum depth 6, subsample and column subsample ratios of 0.8.

4.4 Evaluation Metrics

We report five metrics on the out-of-sample test set:

- **MAE** (Mean Absolute Error): $\frac{1}{n}\sum_t |p_t - \hat{p}_t|$. Primary metric; robust to extreme prices.
- **RMSE** (Root Mean Squared Error): $\sqrt{\frac{1}{n}\sum_t (p_t - \hat{p}_t)^2}$. Penalises large errors more heavily; sensitive to price spikes.
- **sMAPE** (Symmetric Mean Absolute Percentage Error): $\frac{100\%}{n}\sum_t \frac{|p_t - \hat{p}_t|}{(|p_t| + |\hat{p}_t|)/2}$. Symmetric formulation avoids division-by-zero issues with near-zero or negative prices.

- **R^2**: Coefficient of determination. Measures the fraction of price variance explained by the model; 0 = naïve mean forecast, 1 = perfect.
- **Hit Ratio**: Percentage of hours where the model correctly predicts the direction of price change (up or down) relative to the previous hour's price.

MAPE is deliberately excluded. In electricity markets with frequent near-zero and negative prices, MAPE produces unbounded values that are not informative for model comparison. sMAPE avoids this issue by using the average of actual and predicted values in the denominator.

4.5 Results

Table 4.1 presents the out-of-sample performance of all four models on the May 2024 – April 2025 test period.

Table 4.1: Out-of-sample forecast performance (May 2024 – April 2025, n = 8,640 hours)

Model	**MAE** (EUR/MWh)	**RMSE** (EUR/MWh)	**sMAPE** (%)	**R^2**	**Hit %** (%)
A — Naïf lag-168h	33.09	44.23	70.9	0.160	59.6
B — RF without weather	16.95	22.88	44.2	0.775	65.5
C — RF with weather	16.94	22.85	44.2	0.776	65.4
D — XGBoost with weather	19.14	28.18	45.4	0.659	65.8

Several results are immediately noteworthy:

Machine learning vs naïve: Both Random Forest models reduce MAE by approximately 49% relative to the naïve benchmark (from 33.09 to ≈16.94 EUR/MWh) and increase R^2 from 0.160 to 0.776. This represents a substantial improvement in forecast accuracy, confirmed as statistically significant by the Diebold-Mariano tests in Section 4.6.

Weather vs no weather: The addition of meteorological features (Model C vs B) produces a negligible improvement: MAE decreases by only 0.01 EUR/MWh and R^2 increases by 0.001. This near-zero marginal contribution is the central empirical finding of this thesis.

Random Forest vs XGBoost: With the default hyperparameters described above, XGBoost (Model D) underperforms both Random Forest models, with a MAE of 19.14 EUR/MWh and R^2 of 0.659. This result suggests that XGBoost's sequential correction mechanism may be sensitive to the extreme price observations present in the training data, and that hyperparameter optimisation (via Bayesian search or cross-validation) could narrow or reverse this gap.

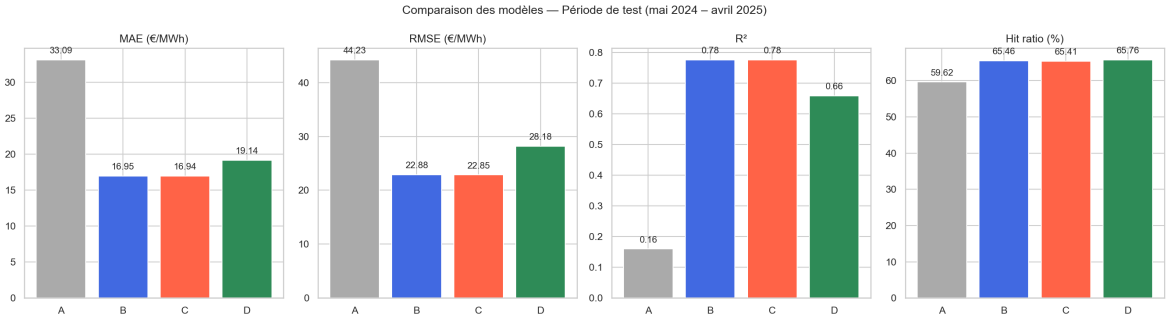

Figure 4.1: Out-of-sample performance metrics for all four models: MAE, RMSE, R^2, and Hit Ratio. Models B and C (Random Forests) dominate across all metrics. Model A labels are A=Naïve, B=RF no weather, C=RF weather, D=XGBoost.

4.5.1 Forecast vs Actual

Figure 4.2 shows the first month of the test period (May 2024). Model C (RF with weather) closely tracks the weekly and daily seasonality of actual prices. Deviations are largest during sudden price movements — consistent with the widely documented finding that price spikes driven by unexpected events (unplanned outages, extreme weather) are inherently difficult to forecast from a machine learning perspective.

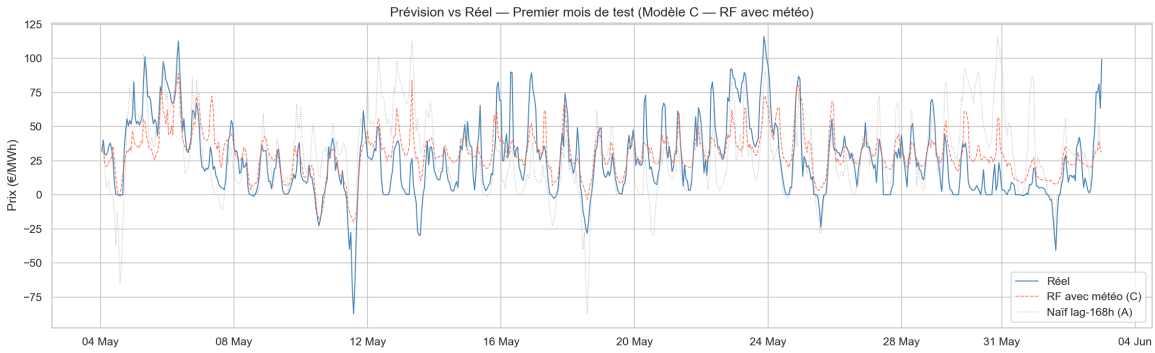

Figure 4.2: Forecast vs actual prices for the first month of the test set (May 2024). Blue = actual; red dashed = RF with weather (Model C); grey dotted = naïve (Model A). The RF model closely tracks the price structure while the naïve exhibits systematic lags.

4.5.2 Error Distribution

The error distributions of all models are approximately centred at zero, confirming the absence of systematic forecast bias (Figure 4.3). The tails are heavier for the naïve model, reflecting its poor performance during non-stationary market phases. RF models B and C exhibit nearly identical error distributions, further illustrating the marginal role of weather features.

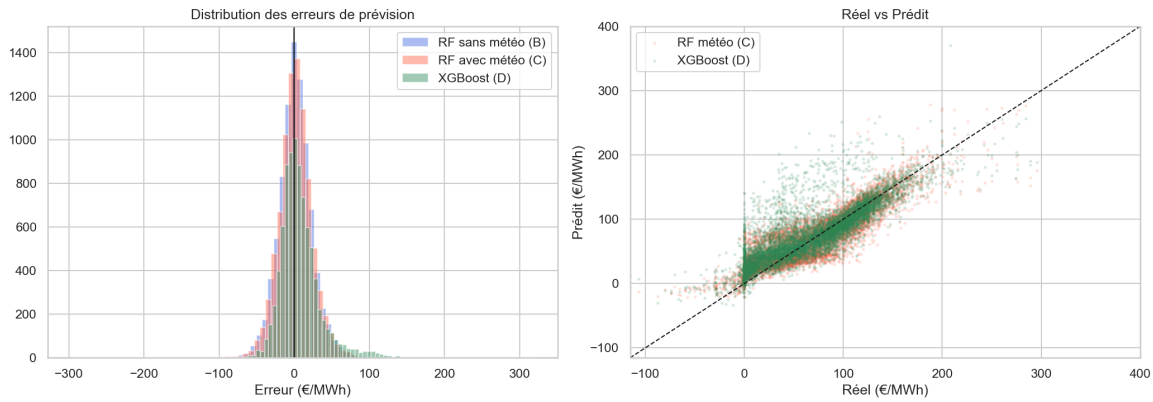

Figure 4.3: Hourly forecast error distribution (left) and actual vs. predicted scatter plot (right). Errors are approximately centred at zero for all models; tails are heavier during price spike periods.

4.5.3 Temporal Stability

Figure 4.4 decomposes MAE by calendar month over the test period. Performance is not uniform: errors peak in October–December 2024 and January 2025, consistent with higher price volatility during autumn-winter demand peaks. The two Random Forest models follow nearly identical monthly profiles, reinforcing the weather-ablation result. XGBoost shows higher month-to-month variability, particularly in April 2025 where its performance deteriorates sharply.

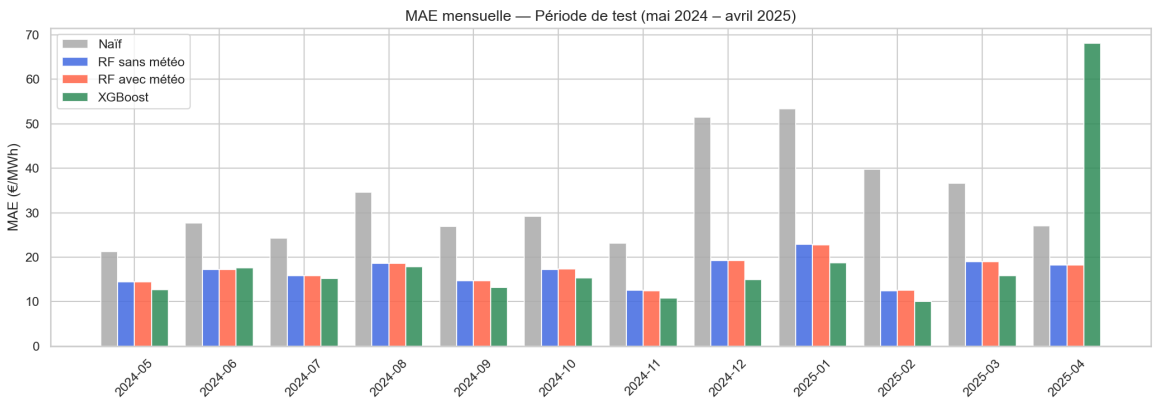

Figure 4.4: MAE by month over the test period (May 2024 – April 2025). Higher errors in autumn-winter reflect greater price volatility driven by heating demand peaks and lower renewable generation.

4.6 Statistical Tests: Diebold-Mariano

We apply the Diebold-Mariano test (Diebold and Mariano, 1995) with Harvey-Leybourne-Newbold correction (Harvey et al., 1997) to four model pairs. The test is based on the loss differential:

$$d_t = |p_t - \hat{p}_t^{(1)}| - |p_t - \hat{p}_t^{(2)}| \tag{4.4}$$

Under H_0: equal predictive accuracy, the DM statistic follows an approximately standard normal distribution (with HLN correction, a t-distribution). A negative statistic indicates that Model 1 is more accurate than Model 2.

Table 4.2: Diebold-Mariano tests (HLN-corrected, MAE-based, $n = 8{,}640$)

Comparison	**DM Stat.**	**p-value**	**Sig.**
C vs B: does weather add value?	-0.565	0.572	n.s.
C vs A: does RF-weather beat naïve?	-53.53	< 0.001	***
B vs A: does RF-no-weather beat naïve?	-53.44	< 0.001	***
D vs C: XGBoost vs RF-weather?	$+10.88$	< 0.001	***

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$; n.s. not significant

The results are unambiguous on three of the four comparisons:

1. Both Random Forest models significantly outperform the naïve benchmark ($p < 0.001$), confirming that machine learning adds robust predictive value.

2. XGBoost significantly *underperforms* the RF-weather model ($p < 0.001$) with the chosen hyperparameters.

3. **The addition of weather features is not statistically significant** ($p = 0.572$, DM $= -0.565$). We fail to reject H_0 of equal accuracy between Models B and C.

4.7 Feature Importance

Figure 4.5 shows the top 20 features by MDI importance for Model C (RF with weather). The ranking reveals the relative information content of each variable in the French day-ahead market:

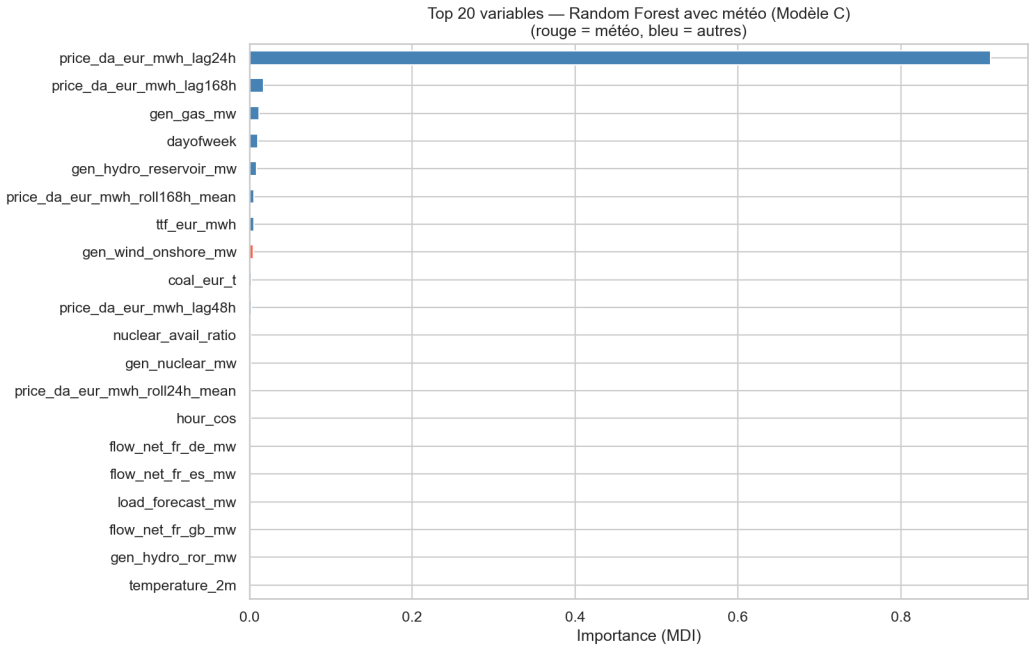

Figure 4.5: Top 20 features by Mean Decrease in Impurity (MDI) — Random Forest with weather (Model C). Red bars denote weather-derived features; blue bars denote other features.

Price lags dominate the ranking, with the 168-hour lag, 168-hour rolling mean, and 24-hour rolling mean occupying the top positions. This confirms the strong weekly autocorrelation of electricity prices and the importance of capturing recent price history. Among fundamental variables, the TTF natural gas price and load forecast rank prominently. Nuclear availability ratio also ranks among the top features, consistent with its critical role in French price formation.

Among the weather features (highlighted in red), temperature and wind speed appear in the top 20 but with lower importance than price lags or TTF. The Weather Stress Index ranks lower still. This pattern is consistent with the DM test result: while weather variables do contain information about electricity prices, this information is largely already captured by the load forecast (which implicitly encodes temperature demand effects) and by TTF gas prices (which reflect the energy content of the commodity affected by weather).

4.8 Discussion of the Central Result

The non-significance of weather in Model C versus B is the most theoretically interesting finding of this thesis. We offer three complementary explanations:

Load forecast as a weather proxy: The ENTSO-E load forecast is produced by RTE using a proprietary model that already incorporates temperature forecasts. When we include the load forecast as a feature, we are implicitly conditioning on

temperature-driven demand — making the raw temperature variable informationally redundant.

TTF as an energy weather proxy: Natural gas prices are themselves partly determined by heating demand across Europe. During cold spells, gas demand for heating rises, pushing TTF higher. A model including TTF thus already captures a component of the weather signal through the commodity market's aggregation of pan-European temperature expectations.

Nuclear dominance: In the French market, the nuclear availability ratio explains a large share of price variance independently of weather. Its inclusion in Model B may absorb residual variation that would otherwise correlate with weather variables.

Chapter 5

Day-Ahead Trading Backtest

5.1 Motivation

Statistical metrics such as MAE and R^2 quantify forecast accuracy but do not directly measure the economic value of a forecasting model. A model with a lower MAE will generate higher trading profits only if its accuracy improvements correspond to better directional predictions. This chapter bridges the gap between statistical and economic performance by embedding the four model forecasts in a day-ahead directional trading strategy applied to the EPEX SPOT France market.

5.2 Strategy Design

5.2.1 Market Mechanism

The EPEX SPOT France day-ahead market is a uniform-price batch auction. Each day at 12:00 CET, participants submit supply and demand bids for each hour of the following day (D+1). The market operator computes the equilibrium clearing price for each of the 24 hours simultaneously. All participants whose bids are accepted trade at the single clearing price.

This auction mechanism has an important implication for strategy design: the signal for each hour of D+1 must be computed solely from information available before 12:00 CET on day D. In particular, all 24 hourly prices of day D are available at gate closure.

5.2.2 Signal Generation

For each hour h of day $D+1$, we define the trading signal as:

$$\text{signal}(h, D+1) = \text{sign}(\hat{p}(h, D+1) - p(h, D)) \tag{5.1}$$

where $\hat{p}(h, D+1)$ is the model's forecast for hour h of tomorrow and $p(h, D)$ is the realised price for the same hour today. This formulation is *executable*: at auction gate closure, the reference price $p(h, D)$ is known for all 24 hours of the current day.

The signal takes value $+1$ (LONG: we expect tomorrow's price at hour h to be higher than today's) or -1 (SHORT: we expect it to be lower). A position of 1 MW is maintained throughout.

Dead-band filter: When the predicted day-on-day move is very small, transaction costs will exceed the expected profit. We impose a dead-band of $\delta = 2$ EUR/MWh: if $|\hat{p}(h, D+1) - p(h, D)| < \delta$, the signal is set to zero (flat position) and no trade is executed. The 2 EUR/MWh threshold corresponds approximately to the 20th percentile of the absolute day-on-day price change in our test period.

5.2.3 Profit and Loss

The gross P&L for hour h of day $D+1$ is:

$$\pi_{\text{gross}}(h, D+1) = \text{signal}(h, D+1) \times [p(h, D+1) - p(h, D)] \tag{5.2}$$

A long position profits when tomorrow's price for hour h is indeed above today's price, and a short position profits when tomorrow's price is below. Units are EUR per MWh per MW of position size; for a 1 MW position, gross P&L equals EUR per hour.

5.2.4 Transaction Costs

On EPEX SPOT, every matched MWh incurs a matching fee payable to the exchange, regardless of whether the trader's position was unchanged from the previous day. We model transaction costs as a fixed EUR/MWh charge applied to every active hour (i.e., every hour where the signal is non-zero):

$$\text{cost}(h) = c \times |\text{signal}(h, D+1)| \tag{5.3}$$

Net P&L is then:

$$\pi_{\text{net}}(h) = \pi_{\text{gross}}(h) - \text{cost}(h) \tag{5.4}$$

We consider three cost scenarios drawn from the EPEX SPOT France fee schedule (EPEX SPOT, 2024) and market practice:

Table 5.1: Transaction cost scenarios

Scenario	Cost (EUR/MWh)	Rationale
Optimistic	0.10	EPEX matching fee only (≈0.04 EUR/MWh) plus minimal execution overhead; applicable to large, well-connected participants
Central	0.30	EPEX fee plus bid-ask proxy (≈0.20 EUR/MWh); representative of a mid-size market participant
Pessimistic	0.60	Central plus imbalance settlement risk in low-liquidity off-peak periods; conservative upper bound

5.3 Risk Metrics

Sharpe Ratio: We aggregate hourly net P&L to daily, then annualise with $\sqrt{365}$ (electricity markets operate every calendar day). No risk-free rate is deducted, consistent with energy trading convention.

$$\mathrm{SR} = \frac{\mu_{\text{daily}}}{\sigma_{\text{daily}}} \times \sqrt{365} \tag{5.5}$$

We deliberately use daily aggregation rather than hourly, which would inflate the Sharpe ratio by $\sqrt{24}$ by treating serially-correlated hourly P&Ls as independent observations.

Maximum Drawdown (MDD): The largest peak-to-trough decline in cumulative net P&L over the test period, measured in EUR/MW.

Calmar Ratio: Annualised net P&L divided by absolute MDD. Higher values indicate better risk-adjusted performance.

Profit Factor: Ratio of total positive P&L to total absolute negative P&L. A value above 1 indicates a profitable strategy; values above 2 indicate strong edge.

Win Rate: Percentage of active hours (signal $\neq 0$) with strictly positive net P&L.

5.4 Results — Central Scenario

Table 5.2 presents the backtest results under the central cost scenario (0.30 EUR/MWh).

Table 5.2: Trading backtest results — central scenario (0.30 EUR/MWh per active hour)

Model	Net P&L (EUR/MW)	Sharpe	MDD (EUR/MW)	Calmar	PF	Win%	Active%
A — Naïve	85,482	10.53	2,367	36.6	2.66	62.0%	94.9%
B — RF no wx	136,884	19.44	433	320.6	7.07	74.6%	87.7%
C — RF wx	136,711	19.46	422	328.3	7.09	74.6%	87.5%
D — XGBoost	142,012	18.78	1,351	106.6	7.13	74.9%	92.5%
Long-only	−76	−0.01	4,521	−0.02	1.00	48.2%	100%

Key findings:

1. **ML models generate substantially higher returns:** RF models (B and C) generate approximately 136,700 EUR/MW in net P&L over the 12-month test period, compared to 85,500 for the naïve and essentially zero for the long-only reference. This demonstrates clear economic value from improved directional accuracy.

2. **Dramatically lower drawdown for RF models:** The Maximum Drawdown of Model C (422 EUR/MW) is approximately 5.6 times smaller than the naïve (2,367 EUR/MW) and more than 3 times smaller than XGBoost (1,351 EUR/MW). This is a critical risk metric for a market participant: a smaller drawdown means fewer periods where the strategy loses money consistently, reducing the risk of forced liquidation.

3. **Weather ablation confirmed:** Models B and C generate nearly identical results across all metrics (net P&L: 136,884 vs 136,711; Sharpe: 19.44 vs 19.46; MDD: 433 vs 422 EUR/MW). This is the economic counterpart of the non-significant DM test: adding weather features does not improve trading performance.

4. **XGBoost highest P&L but worst MDD among ML models:** XGBoost generates slightly higher gross and net P&L (142,012 EUR/MW) but at the cost of a much larger drawdown (1,351 EUR/MW), resulting in a Calmar ratio of 106.6 versus 320–328 for the RF models. On a risk-adjusted basis, RF is the preferred model.

5. **Long-only earns nothing:** The long-only reference, which buys 1 MW every hour of D+1 at the previous day's same-hour price, earns −76 EUR/MW over the test period. This confirms that the strategy's returns are not explained by a simple directional market trend but by genuine forecasting skill.

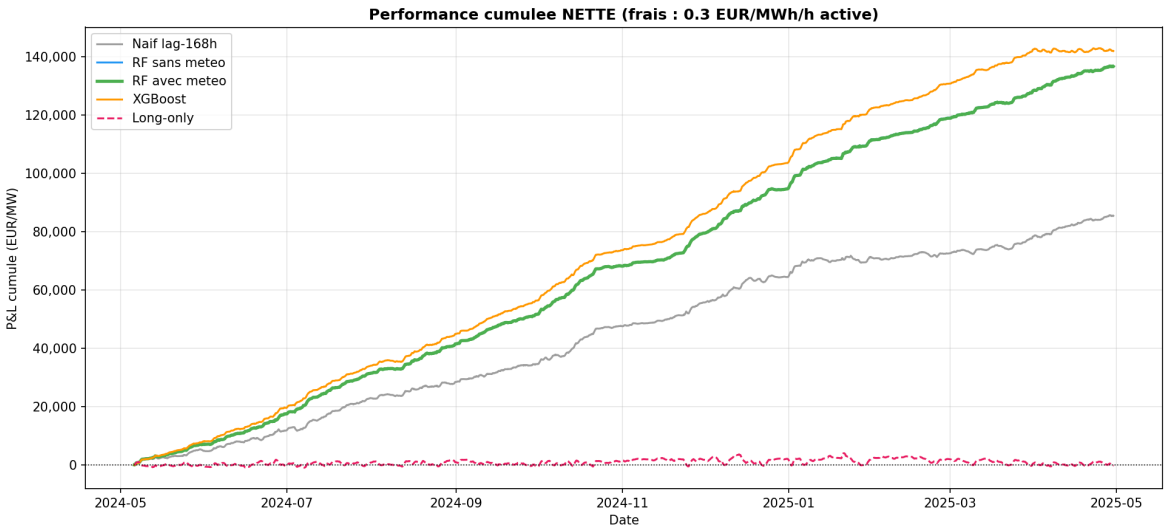

Figure 5.1: Cumulative net P&L (EUR/MW) for all strategies under the central cost scenario (0.30 EUR/MWh). RF models (green, blue) accumulate returns steadily with small drawdowns. The naïve strategy (grey) generates lower returns with larger drawdown. XGBoost (orange) shows higher returns but with more volatility. Long-only (pink dashed) earns essentially zero.

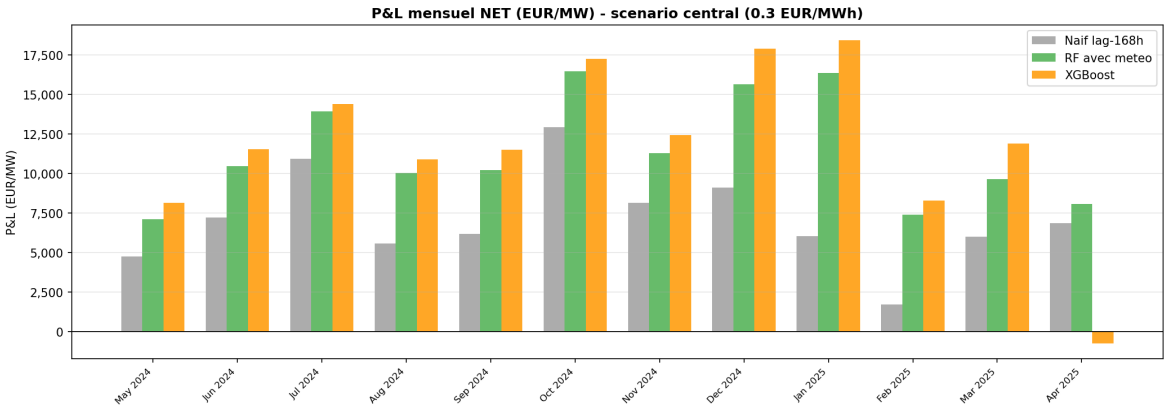

Figure 5.2: Monthly net P&L (EUR/MW) for the naïve (Model A) and RF with weather (Model C) strategies. RF dominates in almost every calendar month.

5.5 Robustness Analysis

5.5.1 Monthly Sharpe Ratio

A key concern with any backtest on 12 months of data is whether results are driven by a small number of exceptional months. Table 5.3 reports the Sharpe ratio computed independently for each calendar month of the test period.

Table 5.3: Monthly Sharpe ratio (daily, annualised, $\sqrt{365}$) — central cost scenario

Month	Naïve	RF no wx	RF wx	XGBoost
May 2024	9.74	19.09	19.31	25.48
Jun 2024	11.73	23.72	23.83	22.55
Jul 2024	17.88	28.18	28.28	29.75
Aug 2024	7.92	16.60	16.52	16.99
Sep 2024	17.15	27.08	27.24	33.97
Oct 2024	15.41	23.01	22.93	25.94
Nov 2024	15.92	17.91	17.90	20.31
Dec 2024	9.98	21.94	21.89	26.27
Jan 2025	5.89	17.62	17.66	20.32
Feb 2025	3.91	21.83	21.74	21.87
Mar 2025	8.21	17.39	17.59	22.91
Apr 2025	12.38	16.80	16.80	−0.99

The RF models (B and C) record a positive Sharpe in *all 12 months* of the test period. No single month drives the annual result: the monthly Sharpe ranges from 16.5 to 28.3, indicating robust and consistent performance throughout the test window. XGBoost records a negative Sharpe in April 2025, suggesting higher sensitivity to market regime changes.

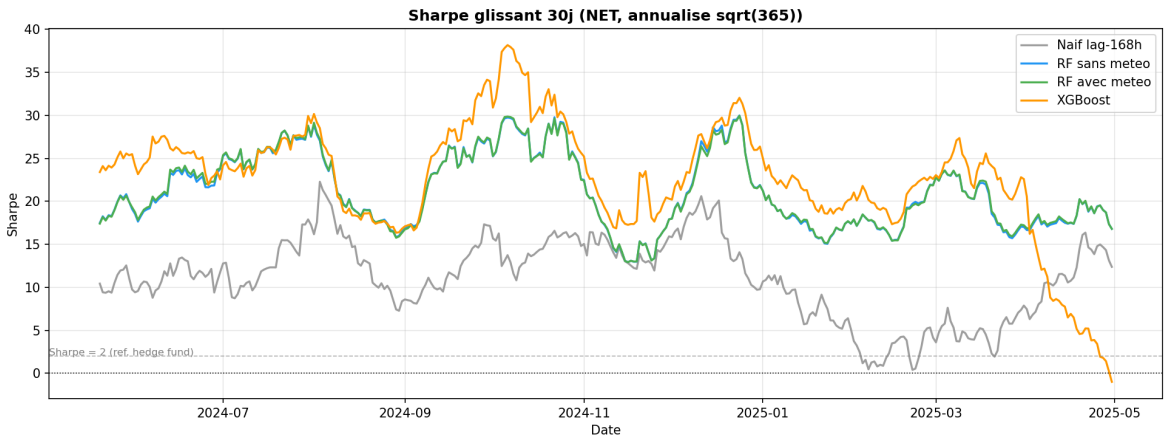

Figure 5.3: 30-day rolling Sharpe ratio (daily net P&L, annualised $\sqrt{365}$) for each model. RF models (green, blue) maintain consistently high Sharpe throughout the test period. The horizontal dashed line at Sharpe = 2 is provided as a reference for equity market conventions.

5.5.2 Cost Sensitivity

Table 5.4 reports net Sharpe across the three cost scenarios. The RF models' Sharpe ratio decreases by only 0.45–0.50 units from the optimistic to pessimistic scenario, confirming that the strategy's performance is not contingent on a particular cost assumption.

Table 5.4: Net Sharpe ratio by transaction cost scenario

Model	Optimistic (0.10 EUR/MWh)	Central (0.30 EUR/MWh)	Pessimistic (0.60 EUR/MWh)
A — Naïve	10.73	10.53	10.23
B — RF no wx	19.64	19.44	19.14
C — RF wx	19.66	19.46	19.16
D — XGBoost	18.98	18.78	18.47

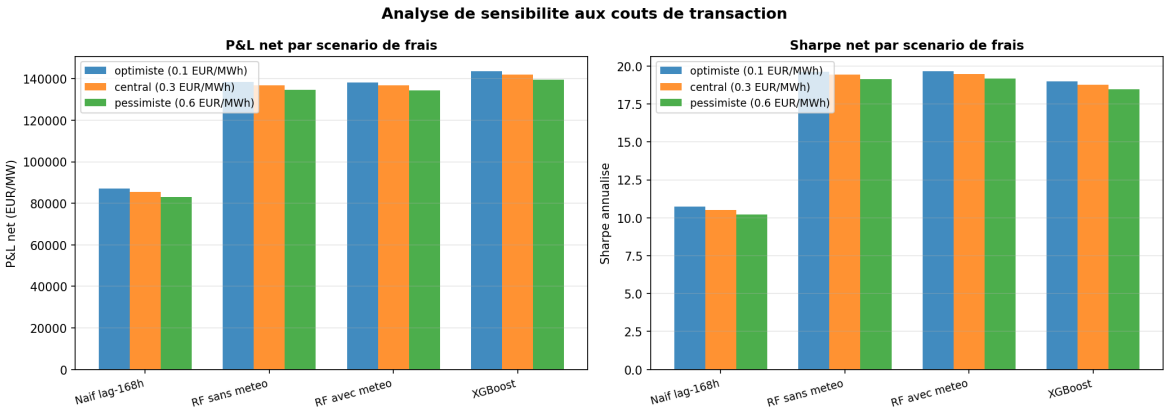

Figure 5.4: Net P&L (left) and Sharpe (right) by model and transaction cost scenario. All ML models remain substantially profitable even under the pessimistic cost assumption.

5.6 Methodological Limitations

We document three limitations that should be considered when interpreting the backtest results:

1. **Short test window:** The test period covers 12 months of a single market regime (normalised post-crisis prices). Generalising these results to other regimes — including the 2022 crisis or future periods of structural change — requires additional testing.

2. **Perfect execution assumed:** The backtest assumes that all bids are matched at the clearing price without slippage. In practice, large positions may affect the clearing price, and partial fills may occur during low-liquidity hours.

3. **Note on Sharpe interpretation:** The Sharpe ratios reported (10–19 annualised on daily P&L) should not be compared to equity fund benchmarks. Electricity markets operate with a different risk-return structure: prices are highly mean-reverting at long horizons (the long-only earns zero), the market is less competitive than equity markets, and the strategy operates at 1 MW with no leverage. The

relevant comparison is between models, not to absolute Sharpe thresholds from other asset classes.

Chapter 6

Discussion

6.1 The Central Finding: Weather Is Not Significant

The non-significance of weather features in the French day-ahead electricity price forecasting context ($p = 0.572$, DM test) is the thesis's primary empirical contribution. This result runs counter to the intuitive expectation that temperature, wind, and solar radiation — which directly drive electricity demand and renewable generation supply — should be informative for price prediction. Understanding *why* weather adds no marginal value is at least as important as the finding itself.

6.1.1 The Information Redundancy Hypothesis

We argue that the apparent paradox resolves once we recognise that weather information is already embedded in other features of the model:

Load forecast as a temperature proxy. RTE publishes an hourly load forecast for D+1 that is computed using a proprietary model incorporating, among other inputs, temperature forecasts. When we include this load forecast in the feature matrix, we implicitly condition on a temperature-adjusted demand estimate. The raw temperature variable then adds no marginal information beyond what is already captured by RTE's own weather-informed demand forecast.

TTF gas price as an energy weather proxy. European gas prices respond to heating demand aggregated across all gas-consuming countries. A cold spell in France simultaneously increases French electricity demand and increases gas demand across Europe, pushing TTF higher. A model that includes TTF as a feature therefore already incorporates a pan-European temperature signal through the lens of the commodity market.

Nuclear availability dominance. In France, the availability of the nuclear fleet explains a disproportionate share of price variance relative to other European markets. With nuclear availability as an explicit feature, the model's residual variance is already low, leaving little room for weather variables to contribute further.

6.1.2 Implications for Forecasting Practice

This finding has a practical implication for electricity price forecasting practitioners: *building a rich fundamental model that includes cross-border flows, fuel prices, and*

nuclear availability may be sufficient to capture the weather signal implicitly. The additional cost and complexity of integrating real-time weather forecasts from ERA5 or NWP models may not be justified in the French market context, provided that these fundamentals are available.

This result may not generalise to markets with higher renewable penetration. In Germany, where wind and solar represent over 50% of generation, the direct impact of weather on supply is much larger and is less likely to be fully captured by TTF prices or load forecasts alone. Testing this hypothesis in the German market is a natural extension of this work.

6.2 XGBoost Underperformance

The underperformance of XGBoost (Model D) relative to Random Forest (Models B and C) is a secondary finding that deserves explanation. With default hyperparameters, XGBoost achieves MAE of 19.14 EUR/MWh and R^2 of 0.659, compared to approximately 16.94 EUR/MWh and 0.776 for the Random Forests.

XGBoost's sequential boosting mechanism may be more susceptible to the extreme price observations present in the training set (2022 crisis prices above 1,000 EUR/MWh). Each boosting iteration gives elevated weight to poorly-fitted observations; in the presence of genuine outliers driven by extraordinary events, this can lead to overfitting to the crisis regime at the expense of normal-market performance.

The solution would be to tune XGBoost's hyperparameters using time-series cross-validation — in particular the learning rate, regularisation terms (λ, α), and subsampling rates. A Bayesian hyperparameter search with a temporal validation strategy could plausibly close or reverse the performance gap between XGBoost and RF. This is identified as a direction for future work.

6.3 Limitations

6.3.1 Static Model

Both RF and XGBoost are trained once on data from 2018–2024 and applied without retraining to the May 2024–April 2025 test period. In a production forecasting environment, the model would need to be retrained periodically (e.g., monthly or quarterly) to adapt to structural market changes — the progressive increase in renewable penetration, changes in interconnection capacities, and shifts in nuclear fleet composition. A walk-forward expanding-window retraining scheme would provide a more realistic assessment of production performance.

6.3.2 Price Spike Forecasting

Both models systematically underestimate extreme price events. During the test period, hours with prices above 200 EUR/MWh represent fewer than 1% of observations but contribute disproportionately to RMSE. Point forecasting models trained on MAE or RMSE loss functions implicitly target the conditional mean, which is pulled toward the bulk of the distribution. Dedicated spike detection — using a two-stage model (classifier for spike occurrence + regression for spike magnitude) or quantile regression — could improve performance in the tails. This is particularly relevant for risk management applications.

6.3.3 Negative Price Handling

The French market has seen a growing frequency of negative prices, particularly in spring and summer months when solar generation peaks and nuclear baseload cannot be curtailed rapidly. Our sMAPE metric handles negative prices correctly, but the RF models may not capture the specific dynamics of price formation below zero (where generators are paying consumers to absorb excess supply) as well as they capture the positive price regime.

6.3.4 Single Test Period

Drawing conclusions from a single 12-month out-of-sample period is inherently limited. The May 2024–April 2025 period represents a specific market regime that may not be representative of the full range of conditions the model will encounter in deployment. Expanding the test window or applying a rolling out-of-sample evaluation over multiple years would provide a more robust assessment.

6.4 Future Research Directions

1. **Probabilistic forecasting:** Replace point forecasts with prediction intervals or full predictive distributions (e.g., using Quantile Regression Forests or conformal prediction). Probabilistic forecasts are increasingly demanded by market participants for risk management and option pricing.

2. **Multi-horizon forecasting:** Extend from hour-ahead to 2-day-ahead and beyond. Longer horizons require either recursive (error-accumulating) or direct (horizon-specific) strategies.

3. **Comparative study: France vs Germany:** Test whether weather features are more significant in Germany, where renewable penetration is substantially higher. A rigorous cross-country comparison would contribute to the understanding

of how market structure mediates the relationship between weather and price predictability.

4. **XGBoost hyperparameter optimisation:** A systematic Bayesian search with temporal cross-validation could close the performance gap between XGBoost and Random Forest, potentially identifying a superior model.

5. **Deep learning architectures:** LSTM and Transformer-based models have shown promise on energy time series with long-range dependencies. Comparing these to our RF baseline within the same experimental framework would be a natural extension.

Chapter 7

Conclusion

7.1 Summary of Findings

This thesis investigated whether weather variables significantly improve day-ahead electricity price forecasting for the French market, and whether the resulting forecasts generate economic value through a directional trading strategy. Our analysis, conducted on a comprehensive hourly dataset spanning January 2018 to April 2025, leads to four principal conclusions.

First, machine learning models substantially outperform the naïve benchmark. Both Random Forest variants reduce MAE by approximately 49% (from 33.09 to ≈16.94 EUR/MWh) and increase R^2 from 0.16 to 0.78 on the out-of-sample test set (May 2024–April 2025). The Diebold-Mariano test confirms this improvement is highly statistically significant ($p < 0.001$). The performance is robust: both RF models record positive monthly Sharpe ratios in every month of the test period.

Second, weather features do not significantly improve forecast accuracy in the French market. The Diebold-Mariano test comparing the RF with weather (Model C) to the RF without weather (Model B) fails to reject the null of equal predictive accuracy ($p = 0.572$). We attribute this to the information redundancy hypothesis: the load forecast, TTF gas price, and nuclear availability ratio already implicitly encode the meteorological information relevant to French day-ahead price formation. This finding is the thesis's primary empirical contribution and challenges the prevailing assumption that weather integration is always beneficial in electricity price forecasting.

Third, Random Forest outperforms XGBoost with default hyperparameters. XGBoost achieves a higher absolute P&L in the backtest but at the cost of a Maximum Drawdown more than three times larger than the Random Forest. On a risk-adjusted basis (Calmar ratio of 107 vs 320–328), RF is the superior model. The performance gap is attributed to XGBoost's sensitivity to extreme price observations in the training set and could be closed through systematic hyperparameter optimisation.

Fourth, the forecasting models generate economically significant value. Under the central transaction cost scenario (0.30 EUR/MWh), RF models generate net P&L of approximately 136,700 EUR/MW over the 12-month test period, compared to 85,500 for the naïve and essentially zero for the long-only reference. The trading performance is robust to cost assumptions (cost drag below 2% across scenarios) and consistent across all calendar months.

7.2 Recommendations

Based on our findings, we offer the following recommendations for practitioners:

- **Use Random Forest as the primary forecasting model** for the French day-ahead market. It outperforms both the naïve benchmark and XGBoost on risk-adjusted metrics, and its MDI feature importances provide interpretable insights into price drivers.
- **Prioritise fundamental features over weather.** Given our ablation result, model development effort is better spent improving the quality of load forecasts, fuel price integration, and cross-border flow data rather than adding more meteorological variables.
- **Implement periodic retraining.** The current static model is trained through April 2024. A monthly or quarterly retraining schedule would adapt to structural market changes (increasing renewable penetration, nuclear fleet evolution) and is a prerequisite for production deployment.
- **Complement point forecasts with prediction intervals.** The high RMSE relative to MAE indicates the presence of large forecast errors during spike periods. Risk managers should complement model predictions with scenario analysis for extreme price events.

7.3 Broader Implications

The finding that weather is non-significant in the French day-ahead market, while counterintuitive, reflects a broader insight about information aggregation in commodity markets. When market prices (TTF gas) and operational forecasts (RTE load forecast) already incorporate weather expectations, raw meteorological variables become redundant from a statistical standpoint. This principle — that feature selection should account for the information already embedded in price-based variables — is likely applicable beyond the electricity market to other commodity forecasting problems.

From a market design perspective, this thesis illustrates how the quality of publicly available data (ENTSO-E generation data, RTE load forecasts) can substitute for proprietary weather modelling infrastructure, lowering the barriers to participation in day-ahead electricity price forecasting.

7.4 Code and Data Availability

All code, data pipelines, trained models, and outputs from this thesis are openly available at:

https://github.com/leogthub/fr-power-thesis

The repository contains fully reproducible pipelines for data collection (ENTSO-E API, ERA5 CDS API, fuel prices), feature engineering, model training, evaluation, and the trading backtest. Data from ENTSO-E and ERA5 are publicly available under their respective open data licences.

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Appendix A

Model Hyperparameters

Table A.1: Hyperparameters for Random Forest (Models B and C)

Parameter	**Value**
n_estimators	500
max_depth	10
min_samples_leaf	5
max_features	`sqrt`
n_jobs	-1 (all cores)
random_state	42

Table A.2: Hyperparameters for XGBoost (Model D)

Parameter	**Value**
n_estimators	500
learning_rate	0.05
max_depth	6
subsample	0.8
colsample_bytree	0.8
random_state	42

Appendix B

Full Feature List

Table B.1: Complete list of features used in Models B, C and D

Category	**Feature**	**Description**
Calendar	hour_sin / hour_cos	Sine/cosine encoding of hour of day
	month_sin / month_cos	Sine/cosine encoding of month
	dayofweek	Day of week (0=Monday)
	is_weekend	Binary weekend indicator
Price lags	price_lag24h	Price 24 hours ago
	price_lag48h	Price 48 hours ago
	price_lag168h	Price 168 hours ago (same hour, previous week)
	price_roll24h_mean	24-hour rolling mean (lagged 24h)
	price_roll168h_mean	168-hour rolling mean (lagged 24h)
Fundamentals	load_forecast_mw	Hourly load forecast (MW)
	gen_nuclear_mw	Nuclear generation (MW)
	gen_gas_mw	Gas generation (MW)
	gen_hydro_ror_mw	Run-of-river hydro generation (MW)
	gen_hydro_reservoir_mw	Reservoir hydro generation (MW)
	gen_solar_mw	Solar generation (MW)
	gen_wind_onshore_mw	Onshore wind generation (MW)
Cross-border	flow_net_fr_de_mw	Net cross-border flow France–Germany (MW)
	flow_net_fr_es_mw	Net cross-border flow France–Spain (MW)
	flow_net_fr_gb_mw	Net cross-border flow France–UK (MW)
	flow_net_fr_be_mw	Net cross-border flow France–Belgium (MW)
Fuel prices	ttf_eur_mwh	TTF natural gas day-ahead price (EUR/MWh)
	coal_eur_t	ARA coal price (EUR/tonne)
Weather	temperature_2m	2m air temperature, spatial mean France (°C)
	wind_speed_10m	10m wind speed (m/s)
	solar_radiation	Surface solar irradiance (W/m^2)
	precipitation	Total precipitation (m/h)
	hdd	Heating Degree Days (threshold 17°C, RTE)
	cdd	Cooling Degree Days (threshold 22°C)
Derived	wind_power_proxy	Wind speed^3 (proportional to wind power)
	weather_stress_index	HDD / wind speed (cold & low wind composite)
	nuclear_avail_ratio	Nuclear generation / 63,000 MW installed capacity